

Differentiable Electrochemical Simulation for Efficient Bayesian Inference



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Abstract

This is a project on the application of gradient computation to the inverse problem in electrochemistry.

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Chapter 1

Introduction

Electrochemistry is a branch of chemistry that is concerned with the relationship between electrical applied potential and chemical change. These reactions arise at the interface between an electronic conductor and an ionic conductor, where charge transfer drives oxidation and reduction of chemical species. The study of these reactions has applications in biochemical sensing, energy storage, technology and many others that underpin much of modern biology and nanotechnology. In particular, electroanalytical techniques have enabled the detection of trace analytes in biological fluids, the development of rechargeable batteries and fuel cells, and the fabrication of nanoscale electronic devices, underlining the broad relevance of electrochemical science across disciplines.

Voltammetry is a common technique to analyse an electroactive species in solution. Voltammetry measures the current that is induced into an electrochemical circuit as the potential of the working electrode is varied with respect to the reference electrode. The current response arises from Faradaic processes, in which electrons are transferred between the electrode surface and the electroactive species, and its magnitude reflects both the kinetics of this charge transfer and the rate at which species are transported to the electrode. Voltammetric techniques are distinguished by how the applied potential is varied over time. In cyclic voltammetry, the potential is swept linearly between two limits and then reversed, producing a characteristic current–potential curve from which thermodynamic and kinetic parameters may be inferred. Other waveforms, such as the sinusoidal perturbations of electrochemical impedance spectroscopy or the staircase profiles of square-wave voltammetry, offer complementary sensitivity and resolution.

Voltammetry has been extensively studied mathematically and numerically. The electron transfer kinetics at the electrode surface are described by the Butler–Volmer

model, which relates the rate of the electrode reaction to the applied overpotential through exponential anodic and cathodic contributions, or alternatively by the Marcus–Hush framework, which grounds the kinetics in a quantum-mechanical treatment of solvent reorganisation. At equilibrium, the Nernst equation dictates the ratio of oxidised to reduced species concentrations at the electrode surface as a function of the applied potential. The transport of electroactive species through the solution is governed by Fick’s laws of diffusion, which, combined with the electrode boundary conditions described above, yield a partial differential equation (PDE) that fully characterises the evolution of the concentration profile in the cell. The solution of this system—the so-called forward problem—predicts the current response given a known set of physical parameters, and has been extensively studied both analytically and numerically [2].

We are interested in the inverse of this process: given the measured current response from a voltammetric experiment, we wish to infer the underlying physical parameters that characterise the electroactive species in solution. These parameters may include the formal potential, diffusion coefficients, and electron transfer rate constants, knowledge of which provides quantitative insight into the thermodynamics and kinetics of the redox system under study. This constitutes the inverse problem, and its solution is considerably more challenging than the forward problem. The mapping from parameters to current is nonlinear and the measurements are corrupted by noise, meaning that naive inversion is typically ill-posed and that uncertainty in the inferred parameters must be carefully quantified.

Many different techniques have been employed to solve this inverse problem. Deterministic optimisation methods, such as least-squares fitting, provide point estimates of the parameters but offer no principled account of uncertainty. Bayesian inference addresses this limitation by treating the parameters as random variables and seeking their posterior distribution conditioned on the observed data. Markov chain Monte Carlo (MCMC) methods have emerged as the dominant computational approach to posterior sampling in electrochemistry, enabling rigorous uncertainty quantification without requiring analytical tractability of the posterior [6]. However, standard MCMC algorithms such as the random-walk Metropolis–Hastings sampler can mix poorly in high-dimensional or geometrically complex parameter spaces, and the requirement to repeatedly solve the forward PDE makes the burn-in phase computationally expensive.

With the advent of modern automatic differentiation (AD) software, it is now possible to efficiently compute exact derivatives of arbitrary numerical programs,

including complete PDE solvers, with respect to their inputs [1]. This capability opens new avenues for gradient-based inference in electrochemistry. In particular, the gradient of the simulated current with respect to the model parameters can be obtained at a cost comparable to a single forward solve, enabling the use of gradient-informed algorithms both during the burn-in phase and throughout the sampling procedure. During burn-in, gradient information can be exploited to drive the chain rapidly toward regions of high posterior probability, dramatically reducing the number of forward solves required before convergence. Once in the posterior, gradient-based samplers such as Hamiltonian Monte Carlo (HMC) use the gradient to make more informed proposals, resulting in higher acceptance rates and faster exploration of the posterior landscape compared with gradient-free alternatives.

This dissertation demonstrates the applicability of gradient-based methodologies to solve the inverse problem in electrochemical experiments. We develop a differentiable PDE solver for common voltammetric waveforms, validate it against known analytical results, and integrate it with gradient-informed MCMC algorithms to perform Bayesian inference of electrochemical parameters. We show that the use of automatic differentiation yields substantial improvements in computational efficiency over gradient-free methods, and we illustrate the approach on both synthetic and experimental data.

Chapter 2

Electrochemical Systems

A typical electrochemical experiment uses an electrochemical cell to drive reduction and oxidation (redox) reactions at the surface of a working electrode. The cell consists of at least two electrodes immersed in an electrolyte solution: the *working electrode*, at whose surface the reaction of interest takes place, and a *counter electrode*, which completes the circuit. A third *reference electrode* is commonly included to provide a stable, known potential against which the potential of the working electrode is measured and controlled. A device called a *potentiostat* maintains the desired potential difference between the working and reference electrodes whilst measuring the current that flows between the working and counter electrodes.

The reaction occurring at the working electrode surface depends on the applied potential and the chemical species present in solution. In the simplest case, a solution-phase species undergoes a one-electron redox reaction:



where species A is reduced — that is, it gains one electron — to form species B. The reverse process, in which B loses an electron to regenerate A, is the corresponding oxidation.

2.1 Voltammetry

Voltammetry is the class of electroanalytical techniques in which the potential applied to the working electrode is varied in a prescribed manner and the resulting current is recorded. Because the current is directly related to the rate of the electrochemical reaction at the electrode surface, the current–potential curve — called a *voltammogram* — encodes information about the kinetics of the redox process as well as the transport of chemical species in solution.

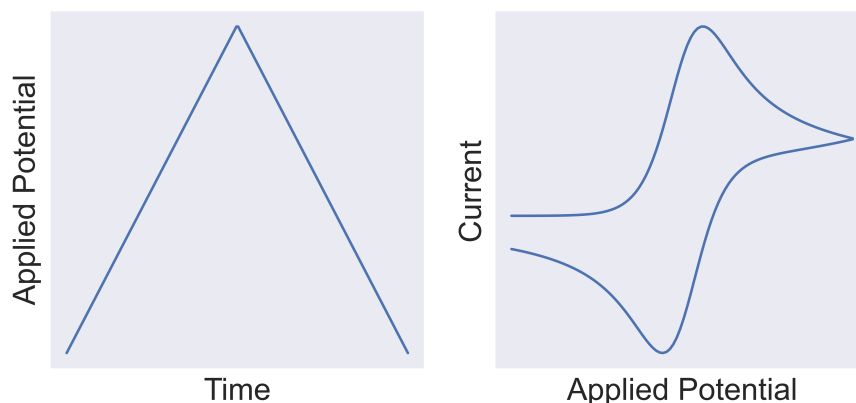


Figure 2.1: Left: the triangular potential waveform applied in a cyclic voltammetry experiment. Right: the resulting reversible reaction’s voltammogram showing the cathodic (reduction) peak which occurs on the forward sweep and the anodic (oxidation) peak which occurs on the reverse sweep.

In a cyclic voltammetry (CV) experiment the applied potential is swept linearly with time from some starting potential E_i , at which the species A is stable and no significant reaction occurs, to a vertex potential E_v , at which electron transfer occurs rapidly to form species B. The potential is then swept back to E_i , causing electron transfer in the opposite direction and the reformation of A. This triangular waveform is illustrated in the left panel of Fig. 2.1.

Throughout this process the current I is recorded. Plotting the current against the applied potential gives the voltammogram shown in the right panel of Fig. 2.1. The forward sweep produces a reduction (cathodic) peak as the potential moves through the range at which A is converted to B; the return sweep produces an oxidation (anodic) peak as B is reconverted to A. The peak shape arises from competition between the electrode kinetics, which accelerate as the driving potential increases, and the depletion of A near the electrode surface caused by the ongoing reaction.

This technique is particularly useful because the position, shape, and relative heights of the peaks provide a signature of the electrochemical process. The potential at which each peak occurs is related to the formal potential E_f° , and the separation between the cathodic and anodic peaks carries information about the electrode kinetics, as discussed in Section 2.2.2.

The *scan rate* ν (V s^{-1}) is the constant rate at which the potential is swept, defined

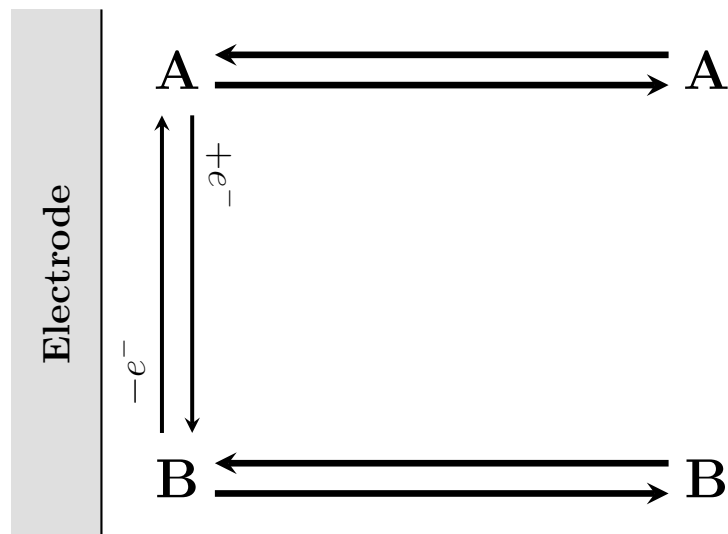


Figure 2.2: Schematic of the one-dimensional electrode reaction. Species A diffuses from the bulk solution towards the electrode surface, where it is reduced ($+e^-$) to form species B. The reverse oxidation ($-e^-$) converts B back to A. Species B diffuses away from the electrode into the bulk.

as

$$\nu = \frac{dE}{dt}. \quad (2.2)$$

The scan rate is an important experimental parameter: faster scans probe shorter timescales and can resolve kinetic processes that would otherwise reach equilibrium on a slower scan, whilst slower scans allow more time for diffusion and can reveal coupled chemical reactions.

2.2 Mathematical Description

The first assumption we make in describing the system mathematically is that we have a “macro”-electrode. This means that the characteristic dimension of the electrode surface is much larger than the distance over which species can diffuse on the timescale of the experiment. As a consequence, we can reduce the dimensionality of the problem and model only the diffusion that acts perpendicular to the electrode surface.

2.2.1 Diffusion: Fick’s Second Law

We model the transport of species in quiescent (unstirred) solution using Fick’s second law [3], which describes the evolution of a concentration field $c(x, t)$ due to diffusion — the net movement of molecules down concentration gradients driven by random

thermal motion. For the couple A/B on a planar electrode we have the system

$$\begin{aligned}\frac{\partial c_A}{\partial t} &= D_A \frac{\partial^2 c_A}{\partial x^2}, \\ \frac{\partial c_B}{\partial t} &= D_B \frac{\partial^2 c_B}{\partial x^2},\end{aligned}\tag{2.3}$$

where D_A and D_B ($\text{m}^2 \text{s}^{-1}$) are the diffusion coefficients of the respective species and x denotes the distance from the electrode surface. Each equation is a linear parabolic partial differential equation defined on the semi-infinite domain $x > 0$.

Given this description of diffusion we are able to reason through the shape of the voltammogram that we observe.

2.2.2 Fick's First Law and Electrode Kinetics

Fick's first law [3] states that the diffusive flux through a plane normal to the x -axis at a point x^* is proportional to the concentration gradient at that point:

$$j_{x^*} = -D \left(\frac{\partial c}{\partial x} \right) \Big|_{x=x^*},\tag{2.4}$$

where the negative sign indicates that the flux is directed down the concentration gradient.

The rate at which the electrode reaction Eq. (2.1) proceeds is governed by the applied potential. The standard model for electrode kinetics is the Butler–Volmer equation [4], which relates the applied potential $E(t)$ to the net flux of A at the electrode surface:

$$j_{0,A} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{RT} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{RT} (E(t) - E_f^0)} \right],\tag{2.5}$$

where k_0 (m s^{-1}) is the standard rate constant, $\alpha \in (0, 1)$ is the transfer coefficient (often taken to be 0.5 for a symmetric energy barrier), F is Faraday's constant, R is the universal gas constant, and T is the absolute temperature. The quantity $E(t) - E_f^0$ is the overpotential, measuring the deviation of the applied potential from the formal potential. The first exponential term represents the reduction contribution: a negative overpotential ($E(t) < E_f^0$) enhances this term and drives the conversion of A to B. The second exponential represents the oxidation contribution, which dominates when the overpotential is positive. In the limit of large k_0 the exponential terms force the surface concentrations towards thermodynamic equilibrium and the system exhibits reversible behaviour; when k_0 is small the reaction is kinetically limited and the flux is directly proportional to k_0 , giving irreversible behaviour.

Combining Eqs. (2.4) and (2.5) at the electrode surface ($x = 0$) yields a Neumann boundary condition for the diffusion equations:

$$-D_A \left(\frac{\partial c_A}{\partial x} \right) \Big|_{x=0} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{RT} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{RT} (E(t) - E_f^0)} \right]. \quad (2.6)$$

Conservation of mass at the electrode surface requires that whatever is consumed of A is produced as B, and vice versa:

$$D_B \left(\frac{\partial c_B}{\partial x} \right) \Big|_{x=0} = -D_A \left(\frac{\partial c_A}{\partial x} \right) \Big|_{x=0}. \quad (2.7)$$

2.2.3 Boundary and Initial Conditions

At the start of the experiment we assume that A is at some bulk concentration c_A^* and that B is at bulk concentration c_B^* :

$$\begin{aligned} c_A(0, x) &= c_A^*, \\ c_B(0, x) &= c_B^*. \end{aligned} \quad (2.8)$$

Far from the electrode surface, the redox reaction has no effect on the concentrations, so the bulk values are recovered:

$$\begin{aligned} c_A(t, x) &\rightarrow c_A^* \quad \text{as } x \rightarrow \infty, \\ c_B(t, x) &\rightarrow c_B^* \quad \text{as } x \rightarrow \infty. \end{aligned} \quad (2.9)$$

2.2.4 Current

The primary experimental observable of a voltammetry experiment is the current measured by the potentiostat. At any instant the current for the reduction of species A is

$$I = FA j_{0,A}, \quad (2.10)$$

where A is the area of the electrode and $j_{0,A}$ is the net flux of species A at the electrode surface, which is proportional to the concentration gradient at that point by Fick's first law Eq. (2.4). Computing the current therefore reduces to evaluating how $j_{0,A}$ evolves in time.

2.3 Non-Dimensionalisation

The combination of Eqs. (2.3) and (2.6) to (2.9) constitutes an initial-boundary value problem. It is far more useful in an electrochemical setting to solve a non-dimensional

version of this problem rather than a particular dimensional instance: the results of a single non-dimensional simulation can be rescaled to any combination of physical parameters without rerunning the computation.

We present a summary of the normalisation in Table 2.1.

Parameter	Normalisation
concentration	$C_j = c_j/c_A^*$
diffusion coefficient	$d_j = D_j/D_A$
spatial coordinate	$X = x/\epsilon$
time	$T = D_A t/\epsilon^2$
applied potential	$\theta = (F/\mathcal{RT}) E$
formal potential	$\theta_f = (F/\mathcal{RT}) E_f^0$
scan rate	$\sigma = (\epsilon^2/D_A)(F/\mathcal{RT}) \nu$
current	$J = I/(\pi \epsilon F D_A c_A^*)$

Table 2.1: Dimensionless normalisation of variables.

We define the dimensionless concentration, diffusion coefficient, spatial coordinate, and time as

$$C_j = \frac{c_j}{c_A^*}, \quad d_j = \frac{D_j}{D_A}, \quad X = \frac{x}{\epsilon}, \quad T = \frac{D_A t}{\epsilon^2}. \quad (2.11)$$

We now substitute these into the governing equations, initial conditions, and boundary conditions.

2.3.1 Governing Equation

Considering the left-hand side of Eq. (2.3):

$$\frac{\partial c_j}{\partial t} = \frac{\partial c_j}{\partial C_j} \frac{\partial C_j}{\partial t} = c_A^* \frac{\partial C_j}{\partial t} = \frac{c_A^* D_A}{\epsilon^2} \frac{\partial C_j}{\partial T}.$$

Now the right-hand side:

$$D_j \frac{\partial^2 c_j}{\partial x^2} = D_j \frac{c_A^*}{\epsilon^2} \frac{\partial^2 C_j}{\partial X^2} = d_j \frac{c_A^* D_A}{\epsilon^2} \frac{\partial^2 C_j}{\partial X^2}.$$

Equating gives the dimensionless diffusion equation:

$$\frac{\partial C_j}{\partial T} = d_j \frac{\partial^2 C_j}{\partial X^2}. \quad (2.12)$$

2.3.2 Initial and Far-Field Conditions

The initial conditions become

$$\begin{aligned} C_A(0, X) &= 1, \\ C_B(0, X) &= C_B^*. \end{aligned} \quad (2.13)$$

In practice we cannot simulate a semi-infinite domain. Einstein's analysis of Brownian motion [5] shows that the root-mean-square displacement of a diffusing particle is $x_{\text{rms}} = \sqrt{2Dt}$. To ensure the outer boundary is sufficiently remote that it is unaffected by diffusion, we truncate the domain at $X_{\text{max}} = 6\sqrt{T_{\text{sim}}}$ in dimensionless form. The far-field boundary conditions are then

$$\begin{aligned} C_A(T, X_{\text{max}}) &= 1, \\ C_B(T, X_{\text{max}}) &= C_B^*. \end{aligned} \tag{2.14}$$

2.3.3 Electrode Surface Boundary

We define the dimensionless applied and formal potentials as

$$\begin{aligned} \theta(T) &= \frac{F}{\mathcal{RT}} E(t), \\ \theta_f &= \frac{F}{\mathcal{RT}} E_f^0. \end{aligned} \tag{2.15}$$

The Butler–Volmer boundary condition Eq. (2.6) and the mass conservation condition Eq. (2.7) then become

$$\begin{aligned} \left. \frac{\partial C_A}{\partial X} \right|_{X=0} &= K_0 [C_A(T, 0) e^{-\alpha(\theta(T) - \theta_f)} - C_B(T, 0) e^{(1-\alpha)(\theta(T) - \theta_f)}], \\ d_B \left. \frac{\partial C_B}{\partial X} \right|_{X=0} &= - \left. \frac{\partial C_A}{\partial X} \right|_{X=0}. \end{aligned} \tag{2.16}$$

2.3.4 Scan Rate

As defined in Eq. (2.2), the scan rate is the rate of change of potential with respect to time. We therefore define a dimensionless scan rate σ as the rate of change of dimensionless potential with respect to dimensionless time:

$$\sigma = \frac{d\theta}{dT}. \tag{2.17}$$

To relate σ to ν we note that

$$\nu = \frac{\partial E}{\partial t} = \frac{\mathcal{RT}}{F} \frac{\partial \theta}{\partial t} = \frac{\mathcal{RT} D_A}{F \epsilon^2} \frac{\partial \theta}{\partial T},$$

and hence

$$\sigma = \frac{F \epsilon^2}{\mathcal{RT} D_A} \nu. \tag{2.18}$$

2.3.5 Current

Since the model is now in dimensionless units we cannot compute the flux $j_{0,A}$ directly, but we can compute a dimensionless flux and relate the two. We define

$$J_{X^*,j} = -d_j \left(\frac{\partial C_j}{\partial X} \right)_{X^*}. \quad (2.19)$$

To recover the dimensional flux, we note that

$$j_j = -D_j \left(\frac{\partial c_j}{\partial x} \right)_{x^*} = -d_j \frac{c_A^* D_A}{\epsilon} \frac{\partial C_j}{\partial X} = \frac{c_A^* D_A}{\epsilon} J_j.$$

Substituting into Eq. (2.10) with $A = \pi \epsilon^2$ gives

$$I = \pi \epsilon F D_A c_A^* J_{0,A}. \quad (2.20)$$

When we run a simulation we record the dimensionless flux J as a function of the dimensionless potential θ . To obtain the physical voltammogram the results need only be scaled by the values of c_A^* , D_A , and ϵ . Consequently, the results of a single simulation apply to any set of these parameters; the simulation does not need to be rerun when, for example, the bulk concentration of A is changed.

2.4 Equal Diffusion Coefficients

For the remainder of this dissertation we assume that all diffusion coefficients are equal, i.e. $d_B = 1$. This assumption allows us to solve for the concentration of A alone and infer the concentration of B.

From the mass conservation condition Eq. (2.16), when $d_B = 1$ the surface flux condition simplifies to

$$\left. \frac{\partial C_B}{\partial X} \right|_{X=0} = - \left. \frac{\partial C_A}{\partial X} \right|_{X=0}. \quad (2.21)$$

Since the diffusion coefficients are equal and both species satisfy the same diffusion equation, the sum $C_A + C_B$ satisfies

$$\frac{\partial(C_A + C_B)}{\partial T} = \frac{\partial^2(C_A + C_B)}{\partial X^2} \quad (2.22)$$

with homogeneous Neumann conditions at $X = 0$ (by Eq. (2.21)) and constant far-field values. Together with the initial conditions, this implies that $C_A + C_B$ is constant everywhere in space and time. Any decrease in C_A is therefore matched by an equal increase in C_B , and vice versa, so that

$$C_A + C_B = 1 + C_B^*. \quad (2.23)$$

We henceforth drop the subscript and write $C \equiv C_A$, with $C_B = 1 + C_B^* - C$. The electrode boundary condition then reduces to

$$\left. \frac{\partial C}{\partial X} \right|_{X=0} = K_0 \left[C(T, 0) e^{-\alpha(\theta(T) - \theta_f)} - (1 + C_B^* - C(T, 0)) e^{(1-\alpha)(\theta(T) - \theta_f)} \right]. \quad (2.24)$$

Chapter 3

Numerical Methods

3.1 Discretisation in space and time

It is clear from the description of the experiment that the most important points in our space are those that are closest to the electrode surface. This is due to the fact that that is where our Butler–Volmer condition operators are where we are measuring the flux during the experiment. On the other hand, we know that points very far away from the surface will not have a large impact on our simulation. This is the intuition for using an exponentially expanding mesh for the discretisation of the spacial dimension. We will use a uniform discretisation for the temporal discretisation.

3.2 Finite Difference Method

We will use a three-point central differencing approximation for the second partial derivatice of the concentration with respect to space:

$$\frac{C_i^k - C_i^{k-1}}{\Delta T} = \frac{\frac{C_{i+1}^k - C_i^k}{\Delta X_+} - \frac{C_i^k - C_{i-1}^k}{\Delta X_-}}{\frac{1}{2}[\Delta X_+ + \Delta X_-]} \quad (3.1)$$

3.3 Numerical Voltammagram

Let us suppose an electrochemical experiment, as in Equation 2.1, in which at the beginning the concentration of the chemical species, A , is uniform throughout the entire solution and there is no B present. When a positive enough potential is applied, A is converted into B at the working electrode, and therefore in the vicinity of the electrode the concentration of A decreases and that of B increases. Consequently, fresh A diffuses towards the electrode from the bulk solution, and B diffuses away from the electrode to regions where there is no B present yet.

Chapter 4

Markov chain Monte Carlo and the Inverse Problem

4.1 Bayesian Approach

4.2 Metropolis-Hastings

4.3 Random Walk Metropolis-Hastings

4.4 Hamiltonian Monte Carlo

Chapter 5

Electron Transfer Reaction

Chapter 6

Heterogeneous Reaction

Chapter 7

Adsorption Reaction

Chapter 8

Conclusion

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